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ABSTRACT

Objectives: Cooling interventions mitigating heat-related declines in autonomic nervous system (ANS) function are understudied in American football. **Methods:** This study exposed fifty, collegiate male athletes during their 2023 season to two bouts of cold air via air-conditioning prior to (60 mins) and during (5 min) training sessions in a hot and humid climate. Armband monitors, (Warfighter Monitor™, Tiger Tech Solutions, Inc., Miami, FL, USA), were used to continuously measure core temperature (°C) during training and baseline heart rate (HR), HR recovery and HR variability (HRV) 24 hours post training. Two HRV time-domain indices were estimated: root mean square of the standard deviation of the NN interval (rMSSD) and the standard deviation of the NN interval (SDNN). These data (2023) were then compared to data collected throughout the previous season (2022) during which players were not provided a cold-air intervention. As such, the 2022 cohort served as a "control" group. **Results:** Linear regression models no statistically significant associations at the following core temperature thresholds: ≥ 37.0°C (baseline HR: p=0.74; HR recovery: p=0.71, rMSSD: p=0.74; SDNN: p=0.78), ≥ 38.0°C baseline HR: p=0.74; HR Recovery: p=0.74; rMSSD: p=0.68 and SDNN: p=0.89) and ≥ 39.0°C: baseline HR: p=0.75; HR Recovery: p=0.82; rMSSD: p=0.78 and SDNN: p=0.83). Compared to the 2022 season, the athletes spent considerably less time training under the higher core temperature thresholds (≥ 38°C: 19.5 min vs 9.7 min; ≥ 39°C: 10.5 min vs 0.9 min), respectively, despite longer overall training duration, (161.1 min vs 187.4 min). **Conclusions:** Our findings suggest that incorporating bouts of cold air exposure when training in high heat and humidity may mitigate the established heat-related deterioration in ANS function.

Key Words: HEAT STRESS, PRE-COOLING, PER-COOLING, AMERICAN FOOTBALL, NERVOUS SYSTEM, COLLEGIATE, EXERCISE TRAINING

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INTRODUCTION

Exercise-induced heat stress, if prolonged and without reprieve, imposes detrimental effects on physiological functions, many of which are regulated by the autonomic nervous system (ANS) (1–3). Research shows that slight increases in core temperature (> 1°C) may induce ANS deterioration that progressively worsens with exposure time, especially when exercising in a hot and humid environment (4). For American football athletes, maintaining physiologically tolerable and safe core temperatures is exceptionally challenging given their heat-resistant uniforms, larger body masses and long duration practices (~2 to 4 hours) (5–7). Thus, finding cooling interventions that effectively mitigate the precipitous increase in core temperature as previously observed in athletes, is imperative for preventing perturbations to ANS function and exertional heat-related illnesses.

Extensive research previously evaluated the effectiveness of different cooling strategies on exercise-induced heat stress in athletes (8–10). Internal and external cooling methodologies were tested including but not limited to cold water or ice ingestion, menthol spray, peripheral and whole-body cold water or ice water immersion, ice garments, moving air (i.e., fan), and cold air exposure (8). Athletes utilized these methods either independently or in combination at different time points relative to their practice or competition (e.g., pre-cooling, per-cooling, post-cooling) (11). Collectively, the evidence suggests that pre-cooling using the whole-body cold water immersion method (CWI) is most effective in lowering skin and core temperatures, subsequently curtailing accelerated increases in core temperature during training in high heat and humidity. While effective, the CWI method is impractical as it is time- and resource-intensive for large teams like American football (n~80 to 100 athletes). Additionally, few studies exist among non-

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endurance sports and sports that require athletes to wear heat-trapping uniforms. Thus, cooling interventions successfully mitigating large increases in core temperature while training in high heat and humidity and preventing deterioration of ANS function remains unclear for sports like American football (7). Our previous work demonstrated the negative influence of elevated core temperatures on ANS function 24 hours post training. During training, athletes reaching a minimum core temperature of 38.0°C showed deterioration of the ANS, reflected by higher baseline heart rates (HR), slower HR recovery post-acute exercise and lower HR variability.(4)

Many collegiate American football programs, especially those located in hot and humid geographical locations like the southeastern United States, are likely equipped with adequately sized, air-conditioned training facilities. Interestingly, cold air exposure as a cooling strategy, for any sport, is extremely limited in the scientific literature (12). Like CWI, cooling via air-conditioning exposes a large proportion of an athlete's body surface area to cooler air, facilitating reductions in skin and core temperatures. Quite possibly, a cold air intervention prior to and during training may successfully temper the accelerated increase in core temperature, allowing athletes to train longer in high heat and humidity without adversely affecting ANS function and augmenting the risk of heat injuries (13). To our knowledge, no studies previously evaluated the effects of intermittent cold air exposure on ANS function in American football. Therefore, the current study investigated the effects of an intermittent cold air exposure on the association between training duration, core temperature thresholds, and ANS recovery. We hypothesized that the cold air exposure would mitigate the heat-related increase in core temperature, allowing athletes to train longer in high heat and humidity without inducing ANS deterioration 24 hours post training.

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MATERIALS AND METHODS

Study Design

The current study analyzed data from a prospective cohort that examined ANS function of Division I collegiate American football athletes during and 24 hours post exercise training throughout two, 27-week seasons (2022 and 2023). Baseline HR, HR recovery and HR variability were measured as indicators of ANS function and were measured only during the weeks the athletes trained (n=25).

Equity, diversity, and inclusion statement

The author group is gender balanced and consists of junior, mid-career and senior researchers from different disciplines; however, all members of the author group are from one country. Two authors are members of a marginalized community. Our study population included male athletes from different socioeconomic backgrounds and minority groups participating in collegiate American Football academies; thus, findings may not be generalizable to settings with fewer resources.

Subjects

Athletes from one Division I collegiate American football team located in southeast Florida, United States, were recruited to participate in the prospective cohort in 2022 (4) and 2023. Minimal differences in participant demographic or anthropometric metrics were observed. One hundred healthy, male football athletes volunteered to participate in the study during the 2022 (n=50) and 2023 (n=50) seasons, respectively. On average, for each cohort, athletes were 21.4 (±1.4) and 19.9 (±1.4) years old, weighed 103.0 (±20.2) and 105.7 (±22.3) kg, were 187.5

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(±6.6) and 189.1 (±7.0) cm tall, 70.0% and 68.0% non-Hispanic black, 20.0% and 16.0% Caucasian, and 10.0% and 16.0% Hispanic, respectively. The anthropometric profiles of the athletes were similar between the 2022 and 2023 cohort. Percent body fat (14.5% vs 15.2%, p=0.49), fat mass (39.2 lbs. vs 37.0 lbs., p=0.56) and skeletal muscle mass (113.4 lbs. vs 113.9 lbs., p=0.81), were not significantly different between the 2022 and 2023 cohort, respectively. All athletes, prior to the study, were fully informed of its protocols, benefits and risks and voluntarily consented. Informed consent was then obtained from all subjects involved in the study. All study protocols followed the ethical principles defined in the Declaration of Helsinki and were approved by the university's Institutional Review Board (IRB #20191223).

Procedures

Exercise Training Sessions. The American football athletes, in both cohorts, participated in the current study during their 27-week season consisting of two, 4-week summer camps, each separated by a week of rest, one 4-week preseason camp and a 13-week in-season (see Figure 1). The exercise training sessions (n=128) lasted, on average, 161.1 (40.6.) min and 187.4 (43.1) min for the 2022 and 2023 cohorts, respectively. All athletes were exposed to the same training which varied in intensity and exercises performed within and between sessions. Training occurred during football practices, were prescribed by the coaching staff and incorporated strength- and power-focused resistive exercises, short-distance sprint intervals, aerobic training, and agility training. Importantly, the study protocols did not modify any aspect of the prescribed training regimen.

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Historical weather data for the 2022 and 2023 seasons (May through November) were obtained from the National Weather Service (National Atmospheric and Oceanic Administration [NOAA]). Average ambient sea level temperatures in Miami, FL for the 2022 season ranged from 26.1°C to 29.6°C with a minimum and maximum temperature of 22.4°C and 33.2°C, respectively. Average ambient sea level temperatures in Miami, FL for the 2023 season ranged from 25.4°C to 30.0°C with a minimum and maximum temperature of 21.9°C and 34.4°C, respectively. The average relative humidities throughout the 2022 season ranged from 71.1% to 78.9% (min = 38.3%, max=100%). The average relative humidities throughout the 2023 season ranged from 68.7% to 75.6% (min = 29.3%, max=100%).

Cold Air Exposure Intervention. Whole-body cold air exposure was implemented during the 2023 season (May to November) and was used as the cooling intervention method and was incorporated twice during each training session. The cold air was delivered via an air-conditioned, indoor practice facility (90,000 ft^2) on the university's campus. The air conditioning system was set at 15.6°C. The first exposure occurred at the beginning of the training session where athletes performed their warmup routines and sport-specific drills for 45 to 60 minutes. Following, the athletes moved to the outdoor training facility and engaged in 30 to 45 minutes of exercise. A 5-minute break inside the air-conditioned facility occurred immediately after training outdoors. Here, athletes only removed their helmets, then rested, and performed no additional exercises. Athletes then returned outdoors for an additional 30-minute training session.

Physiological Measurements. Throughout each training session, participants wore armband monitors that continuously measured core temperature, HR, and indices of HRV (Warfighter

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Monitor (WFM), Tiger Tech Solutions Inc., Miami, FL, USA). The WFM is equipped with several biometric capabilities including but not limited to temperature sensor and electrocardiography and was previously validated in many subpopulations, including athletes (4, 14, 15). The monitors were secured on the upper left arm with an elastic band encompassing the widest part of the biceps muscle. Monitors were worn prior to and throughout each training session.

Core Temperature. Core body temperature was derived from temperature sensors on the WFM. The temperature sensors on the WFM are medical-grade infrared temperature sensors which have been calibrated to core body temperature across thousands of subjects in a hospital setting utilizing health monitors and Swan–Ganz catheters as the gold standard measure(16, 17). Core temperature was measured throughout the duration of each training session.

24 h ANS Recovery and Function. The current study used electrocardiogram-based metrics to represent ANS recovery and function given that cardiac activity is a strong indicator of ANS activity. The release of acetylcholine and epinephrine from cardiac nerve fibers controls the firing rate of the sinoatrial (SA) node affecting cardiac activity (18, 19). Thus, cardiac metrics like baseline HR, HR recovery and heart rate variability (HRV) provide accurate information on the interaction between parasympathetic and sympathetic nervous systems, the two branches of the ANS.

Baseline HR. Baseline HR represented ANS recovery and was measured in the morning (0600–0700), prior to the start of a football training session and 24 h after the start of the previous day's

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training session. Baseline HR was obtained following at least 4 min of inactivity, per established protocols (20). During this measurement, all athletes remained seated and nearly motionless for 5 min to collect a "resting" baseline HR.

HR Recovery. HR recovery was measured throughout the next day's football training session, 24 h following the previous day's training session. HR recovery was defined as the reduction in HR during 30 s rest intervals and represented localized parasympathetic activation. HR recovery was measured within the first 30 s of the rest interval, as during this period HR exhibits the greatest rate of change following an acute bout of exercise (21). HR recovery was estimated during all rest intervals occurring within a training session and then averaged.

HRV. Like baseline HR, HRV was measured in the morning (0600–0700), prior to the start of a football training session and 24 h after the start of the previous day's training session. Per established protocols (20), HRV was obtained following at least 4 min of inactivity. During this time, all athletes remained seated and nearly motionless for 5 min to collect a "resting" HRV measure. Two HRV time-domain indices were measured which assessed the changes in the inter-beat interval including RR and NN intervals. RR intervals were the time between R waves on consecutive QRS complexes and NN intervals were noise-free RR intervals. From these data, the two separate time-domain indices were derived including SDNN (standard deviation of the NN interval) and RMSSD (the root mean square of the standard deviation of the NN interval). These HRV time-domain indices have been shown to reflect parasympathetic autonomic output (rMSSD and SDNN) and sympathetic output (SDNN) (18) and are more robust to external stimuli like respiration cycles(22, 23).

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Because the current study focused on ANS recovery and function 24 h post-exercise training, baseline HR, HR recovery and HRV were not measured following one or more rest days. For example, baseline HR, HR recovery and HRV were not measured on Mondays during summer training camps as this day was preceded by two rest days (Saturday and Sunday). The inclusion of rest days in the analyses would likely weaken the associations and, thus, less accurately estimate the acute impact of exercise training.

Statistical Analysis

This study investigated the relationships between the time spent over varying core temperature thresholds with and without whole-body cold air exposure prior to and during acute bouts of exercise training and its influence on ANS recovery and function within 24h post-exercise. The cooling method, duration (in minutes) of exercise training and core temperature (°Cserved as the primary independent variables. 24-hr HR and HRV indices (SDNN and rMSSD), and HR recovery served as the primary outcome variables and represented ANS recovery and function within 24h post-exercise training, respectively. The associations between exercise training duration and ANS recovery and function were stratified by data driven core temperature thresholds: ≥37°C, ≥38°C, and ≥39°C. The conditional distributions for each association were tested for normality using the Kolmogrov-Smirnov test and found to be normally distributed. Linear regression models were used to quantify these associations and were performed separately for each temperature threshold and ANS metric. Because these analyses were previously performed and published for the 2022 season, only data from the 2023 season were analyzed in the current study. The published 2022 data are shown for comparative purposes. Differences in demographic and anthropometrics characteristics between the 2022 and

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2023 cohorts were tested using Pearson's studentized t-tests and Chi-square test for independence. For all models, β coefficients and standard errors were estimated, and the a priori threshold for statistical significance was set at $\alpha = 0.05$. Statistical analyses were performed in MATLAB, version 2021b (MathWorks, Natick, MA).

RESULTS

Table 1 presents details on the average duration of exercise training sessions, the time athletes spent under specific core temperature thresholds, and their ANS recovery and function 24 hours post exercise training. Across the 2023 season, the athletes participated in 128 training sessions that, on average, lasted 187.4 (± 43.1) minutes. Athletes in the 2023 cohort appeared to spend less time exercise training under the higher core temperature thresholds (≥38 °C: 9.7 vs 19.5 min and ≥39 °C: 0.9 vs 10.5 min). Baseline HR and HR recovery appeared similar between the 2022 and 2023 cohort (61.4 vs 62.9 bpm and 30.6 vs 31.3 bpm, respectively. The differences in HRV 24 hours post exercise training, however, appeared larger with higher rMSSD (82.1 vs 72.0 ms) and SDNN (125.6 vs 106.7 ms) values for the 2023 cohort.

The associations between duration of exercise training sessions and ANS recovery and function, evaluated under three separate core temperature thresholds are presented in tables 2 and 3. For the ≥37°C threshold, both cohorts (2022 vs 2023) showed null associations between duration of exercise training and ANS recovery and function (baseline HR: $R^2 = 0.03$, p=0.698 vs 0.03, p=0.738; HR recovery: $R^2 = 0.02$, p=0.609 vs 0.02, p=0.709; rMSSD: $R^2 = 0.04$, p=0.635 vs 0.05, p=0.743 ; SDNN: $R^2 = 0.03$, p=0.529 vs 0.02, p=0.776, respectively). Conversely, for the 2023 cohort, the associations between duration of exercise training and ANS

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recovery and function were nullified for the ≥38°C and ≥39°C thresholds. For the 2022 cohort, however, these same associations were statistically significant and showed larger, meaningful magnitudes in the slope and R^2 values.

DISCUSSION

The current study investigated the effects of two intermittent cooling sessions via cold air exposure on the association between training duration, core temperature thresholds, and ANS recovery. We hypothesized that the cold air exposure would mitigate the heat-related increase in core temperature, allowing athletes to train longer in high heat and humidity outdoors without inducing ANS deterioration 24 hours post training. The major finding of this study was that athletes enduring a cold air exposure prior to and during exercise training in high heat and humidity nullified the association between training duration, elevated core temperatures, and ANS deterioration 24 hours post-exercise training. The null relationships observed were likely consequent to athletes spending considerably less time training under the higher core temperature thresholds (≥ 38°C and ≥ 39°C), allowing athletes to train for longer durations without negatively affecting ANS function the next day. Our observations may suggest that providing athletes intermittent exposures to cold air of sufficient duration may mitigate the rise in core temperature consequent to prolonged exercise training in hot and humid climates and alleviate any associated negative effects on ANS recovery and function.

The most novel aspect of the current study was that the cold air exposure appeared to nullify the association between training duration in high heat and humidity, elevated core temperatures, and 24-hour ANS deterioration. Specifically, athletes exposed to cold air prior to

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and during exercise trainings sessions in a hot and humid climate, did not exhibit decrements in baseline HR, HR recovery or HRV 24 hours post exercise training, despite expected heat-related increases in core temperature. These observations starkly contrast our recently published work that explored the relationships between prolonged exercise training duration in high heat and humidity and ANS recovery and function (see tables 2-3) (4). In that study, the athletes performed similar practice sessions with respect to duration and exercise training regimen. The athletes, however, were not exposed to any cooling method prior to, during or following each training session. Not surprisingly, significant associations between exercise training duration, under specific core temperature thresholds, and ANS recovery and function were observed. At the higher core temperature thresholds (≥ 38°C and ≥ 39°C), athletes experienced higher baseline HRs, slower HR recovery during rest intervals and impaired HRV indicating significant deterioration of the ANS 24 hours post training. Thus, the observation of the current study in addition to our previous study suggest that the cold air exposure method may mitigate the negative influences of prolonged exercise training in high heat and humidity on the ANS (24–26).

The null associations observed may be explained by the athletes spending considerably less time training under the higher core temperature thresholds, despite longer overall training duration, relative to the 2022 cohort (≥ 38°C: 19.5 min vs 9.7 min; ≥ 39°C: 10.5 min vs 0.9 min, respectively). This finding suggests that cold air exposure, of sufficient duration, may augment an athlete's heat storage capacity (9, 27). Current evidence implicates that the gradient between core temperature and skin temperature (core: skin) strongly influences an individual's thermoregulatory response to exercising in the heat. Specifically, several studies previously

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demonstrated cooling methods that effectively lower skin temperature, and ideally core temperature, improve exercise performance (e.g., time-to-exhaustion, maximal capacity). Unfortunately, few studies have evaluated the effectiveness of cold air exposure on skin and core temperatures, yielding inconsistent findings. However, one cooling strategy, whole-body cold-water immersion that is like the cold-air exposure method and extensively studied, effectively lowers skin and core temperatures (27). The cooling of the large skin surface area induces peripheral vasoconstriction, recirculating colder blood back to the thoracic region, lowering core temperature and reducing the thermo-physiological strain of exercise training in high heat and humidity. As such, athletes, as shown in the current study and others (8, 27, 28), may perform exercise in the heat and humidity for longer durations and delaying the onset of heat-related declines in performance. We acknowledge however, that it is also possible that the longer training durations in the 2023 cohort could be attributed to variation in training regimens and accounting for transition times between the cooling phases. During these transitions, athletes (n=100) gathered their gear, regrouped and returned outdoors for additional training.

Unlike several former studies (13, 29), the current study implemented the cooling method prior to and during exercise training sessions. Additionally, during the pre-cooling phase, the athletes were performing low to moderate intensity exercise. This contrasts to many other cooling methods where athletes remain stationary during their pre-cooling exposure which may be disadvantageous when transitioning to exercise training in high heat and humidity (10, 11). This method may result in simultaneous increases in core and skin temperature during the onset of exercise. Consequently, this may accelerate the decrease in the core: skin temperature gradient and increasing core temperature more quickly. The method used in the current study, however,

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potentially mitigated this response as the athletes were exposed to cold air while simultaneously exercising. As such, the transition to exercising outdoors in the high heat and humidity may affect the core: skin temperature gradient to a lesser degree (30, 31) allowing for longer training duration. Moreover, the mid-session cooling phase (~5 min) implemented following 45 to 60 mins of exercise training outdoors likely slowed the increase in core temperature and narrowing of the core: skin temperature gradient. This response allowed athletes to perform an additional bout of outdoor training, staving off the rapid increase in core and skin temperatures. In support, athletes in the current study were able to train for longer (187.4 vs 161.1 min) and spent less time under higher elevated core temperatures. Our finding suggests that implementing both cooling sessions may more effectively mitigate the precipitous increase in core temperature (10, 13) during outdoor training in high heat and humidity.

Practical Implications

Critically, American football ranks in the top 5 for sports with the highest prevalence of exertional heat-related illnesses (3), conditions that negatively impact the ANS. Unfortunately, the evidence base for practical, effective cooling strategies in this sport is extremely limited. As such, identifying that cold air exposure effectively mitigates the precipitous increase in core temperature, possibly preventing deterioration of the ANS. This finding profoundly impacts the scientific literature and sports realm. Moreover, this cooling method is significantly less burdensome compared to other effective methods like cold water immersion and ice garments. These methods are not conducive to sport-specific movement nor practical application for large sports teams like American football which includes ≥ 100 athletes on roster. However, the cold

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air exposure method is time efficient and only requires an air-conditioned facility that likely already exists at most collegiate American football programs.

The current study has several novel aspects that significantly expand the existing evidence base. First, we uniquely identified a cooling strategy for collegiate American football athletes, an understudied subpopulation, that effectively alleviates the previously documented ANS deterioration consequent to prolonged exercise training in a hot and humid climate. Second, the 2022 cohort was similar regarding the demographic and anthropometric athlete profiles, exercise training regimens and climate, allowing for appropriate comparisons between the interventions. Third, core temperature and ANS deterioration were estimated using the WFM which used electrocardiography and infrared temperature technology, both previously validated against their respective gold standard methodologies. Fourth, the study was implemented in a natural setting strengthening the translation of these findings to other sport programs. The current study also has limitations. First, although the comparative cohort was similar, the study did not implement the intervention groups simultaneously, increasing variability in other possible confounding factors. Second, our study design did not include a pre-cooling only or per-cooling only group, as such, conclusions on the independent effect of each cooling interval cannot be drawn. Third, because skin temperature was not measured, we are unable observe any changes in the core: skin temperature gradients. Fourth, this study was conducted in one sample of male, collegiate American football athletes from one geographical location and as such, our observations may not be generalizable to other sports and/or female athletes. Lastly, the hydration and nutrition status of the athletes were not measured.

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CONCLUSIONS

The findings of the current study demonstrate the potential effectiveness of cold air exposure mitigating the precipitous increase in core temperature and subsequent deterioration of the ANS. Specifically, athletes exposed to 45 to 60 minutes of cold air prior to and during a single, 5-minute rest interval while training outdoors in high heat and humidity decreased the time spent training at elevated core temperature thresholds (≥ 38°C and ≥ 39°C). Moreover, the athletes, on average, did not exhibit signs of ANS deterioration unlike our former study. The current study uniquely identified an effective cooling strategy that may help prevent deterioration of the ANS in collegiate, American football athletes, allowing coaching staff to safely train their athletes during seasons with high heat and humidity. We strongly recommend that future studies evaluate the effectiveness of existing cooling intervention methods in American football. Further, for the cold air method, researchers should explore different durations and frequencies to determine the most effective cold air exposure.

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Author Contributions: For research articles with several authors, a short paragraph specifying their individual contributions must be provided. The following statements should be used Conceptualization, H.L.W., S.H.W., M.J.W., P.G., and L.A.F.; methodology, S.H.W., H.L.W., M.J.W., E.R., and T.R.; software, S.H.W, H.L.W.; validation, H.L.W, M.J.W., P.G. and L.A.F.; formal analysis, M.J.W., H.L.W., S.M.M.; investigation, D.H., E.R., T.R., P.G. and L.A.F.; resources, H.L.W., L.A.F., and E.R.; writing—original draft preparation, S.M.M; writing—review and editing, S.M.M., H.L.W., and M.J.W; visualization, M.J.W.; supervision, E.R., L.A.F. and H.L.W.; project administration, P.G., L.A.F., E.R. and H.L.W.; funding acquisition, N/A. All authors have read and agreed to the published version of the manuscript. H.L.W. is the guarantor and accepts full responsibility for this finished work, conduct of the study, analysis of the data and the decision to publish.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

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Data Availability Statement: The data can be provided by Tiger Tech Solutions, Inc. pending scientific review and a completed materials transfer agreement. Requests for the data should be submitted to the corresponding author: Samantha M McDonald, smmcdo4@ilstu.edu.

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Conflicts of Interest: The following authors are paid employees of Tiger Tech Solutions, Inc, the owner of the Warfighter Monitor used in this study: SHW, MJW, HLW, DH, and SMM. Authors ER, TR, PG and LAF report no conflicts of interest.

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FIGURE LEGEND

Figure 1. Prospective study design for exercise training sessions and measurement days.

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Figure 1

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Table 1. Duration of acute exercise training sessions, time spent under specific core temperature thresholds and 24 h ANS recovery and function, by intervention.

	2022 Cohort	**2023 Cohort**
	No Intervention	**Cold Air Intervention**
	Mean (SD)	*Mean (SD)*
No. of Training Sessions	128	128
Duration of Sessions (min)	161.1 (40.6)	187.4 (43.1)*
Duration (min) Above Temperature Thresholds		
≥37 °C	30.4 (35.9)	52.5 (50.9)*
≥38 °C	19.5 (21.5)	9.7 (20.7)*
≥39 °C	10.5 (12.9)	0.9 (5.1)*
24 h ANS Recovery and Function		
Baseline HR (bpm)	61.4 (8.6)	62.9 (7.1)
30 s HR Recovery (bpm)	30.6 (6.0)	31.3 (5.1)
rMSSD (ms)	72.0 (7.0)	82.1 (11.4)*
SDNN (ms)	106.7 (12.2)	125.6 (21.7)*

*denotes statistical significance ($p<0.05$

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Table 2. Adjusted linear regression coefficients for the relationships between time spent under different temperature thresholds and 24 h ANS Recovery and Function.

	2022 Cohort No Intervention				2023 Cohort Cold Air Intervention			
	β (SE)	*Adj. R^2*	*95% CI*	*p-value*	*β (SE)*	*Adj. R^2*	*95% CI*	*p-value*
Baseline HR								
Duration Above Temperature Thresholds								
≥37 °C	0.01 (0.01)	0.03	(−0.02, 0.02)	0.698	0.01 (0.01)	0.03	(−0.01, 0.02)	0.738
≥38 °C	0.10 (0.02)	0.26	(0.06, 0.14)	p < 0.0000	0.03 (0.002)	0.02	(-0.06, 0.04)	0.744
≥39 °C	0.39 (0.05)	0.62	(0.30, 4.49)	p < 0.0000	-0.003 (0.002)	0.01	(-0.01, 0.02)	0.747
HR Recovery								
Duration Above Temperature Thresholds								
≥37 °C	0.003 (0.01)	0.02	(−0.01, 0.02)	0.609	0.01 (0.001)	0.02	(−0.01, 0.03)	0.709
≥38 °C	−0.06 (0.02)	0.21	(−0.10, −0.03)	0.0002	−0.004 (0.002)	0.01	(−0.01, 0.03)	0.739
≥39 °C	−0.26 (0.04)	0.52	(−0.34, −0.18)	p < 0.0000	−0.01 (0.002)	0.02	(-0.02, 0.03)	0.821

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Table 3. Adjusted linear regression coefficients for the relationships between time spent under different temperature thresholds and 24 h heart rate variability.

	2022 Cohort No Intervention				**2023 Cohort** Cold Air Intervention			
	β (SE)	*Adj. R^2*	*95% CI*	*p-value*	*β (SE)*	*Adj. R^2*	*95% CI*	*p-value*
rMSSD								
Duration Under Temperature Thresholds								
≥37 °C	−0.01 (0.01)	0.04	−0.03, 0.02	0.635	-0.02 (0.01)	0.05	(−0.04, 0.02)	0.743
≥38 °C	−0.11 (0.02)	0.24	−0.15, −0.06	p < 0.0000	-0.01 (0.001)	0.01	(-0.06, 0.14)	0.682
≥39 °C	−0.37 (0.05)	0.58	−0.47, −0.26	p < 0.0000	0.03 (0.01)	0.03	(-0.00, 0.05)	0.776
SDNN								
Duration Under Temperature Thresholds								
≥37 °C	−0.01 (0.02)	0.03	−0.05, 0.03	0.529	0.02 (0.01)	0.02	(−0.01, 0.04)	0.776
≥38 °C	−0.16 (0.04)	0.22	−0.24, −0.08	p < 0.0000	−0.06 (0.02)	0.01	(−0.10, 0.03)	0.884
≥39 °C	−0.67 (0.09)	0.58	−0.85, −0.49	p < 0.0000	−0.03 (0.01)	0.01	(−0.27, 0.18)	0.832

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